Matemáticas B
4º ESO

SOLUCIONES DE LA FICHA · UNIDAD 5

Trigonometría

FentMates

Intenta resolver cada ejercicio antes de mirar su solución. Los ángulos se han redondeado al segundo y las longitudes, a la centésima (calculando siempre con los valores sin redondear). Puedes comprobar muchos resultados con las herramientas interactivas de los apuntes de la unidad.

1 Medida de ángulos: grados y radianes

1. ★☆☆

a) $\boxed{\frac{\pi}{4}}$ **b)** $\boxed{\frac{\pi}{6}}$ **c)** $\boxed{\frac{2\pi}{3}}$ **d)** $\boxed{\frac{5\pi}{6}}$ **e)** $\boxed{3\pi}$ **f)** $\boxed{6\pi}$

g) $\boxed{10\pi}$ **h)** $\boxed{\frac{4\pi}{3}}$ **i)** $\boxed{\frac{31\pi}{18}}$ **j)** $\boxed{\frac{7\pi}{18}}$ **k)** $\boxed{\frac{\pi}{15}}$ **l)** $\boxed{4\pi}$

2. ★☆☆

a) $\boxed{180°}$ **b)** $\boxed{720°}$ **c)** $\boxed{1260°}$ **d)** $\boxed{30°}$ **e)** $\boxed{135°}$ **f)** $\boxed{270°}$

g) $\boxed{300°}$ **h)** $\boxed{90°}$ **i)** $\boxed{240°}$ **j)** $\boxed{216°}$ **k)** $\boxed{72°}$ **l)** $\boxed{20°}$

3. ★☆☆

Grados	$30°$	$\boxed{45°}$	$60°$	$\boxed{120°}$	$135°$	$\boxed{150°}$	$210°$	$\boxed{225°}$	$300°$	$\boxed{330°}$
Radianes	$\boxed{\frac{\pi}{6}}$	$\frac{\pi}{4}$	$\boxed{\frac{\pi}{3}}$	$\frac{2\pi}{3}$	$\boxed{\frac{3\pi}{4}}$	$\frac{5\pi}{6}$	$\boxed{\frac{7\pi}{6}}$	$\frac{5\pi}{4}$	$\boxed{\frac{5\pi}{3}}$	$\frac{11\pi}{6}$

4. ★☆☆

a) $\boxed{90° = \frac{\pi}{2}}$ **b)** $\boxed{270° = \frac{3\pi}{2}}$ **c)** $\boxed{60° = \frac{\pi}{3}}$ **d)** $\boxed{120° = \frac{2\pi}{3}}$

5. ★★☆ $\widehat{AOB} = \frac{360°}{6} = \boxed{\frac{\pi}{3}}$; ángulo interior: $\widehat{ABC} = 120° = \boxed{\frac{2\pi}{3}}$; el triángulo ABC es isósceles, y $\widehat{BAC} = \frac{180°-120°}{2} = 30° = \boxed{\frac{\pi}{6}}$; AD es un diámetro y el ángulo inscrito $\widehat{ACD}$ es recto: $\boxed{\frac{\pi}{2}}$.

6. ★★☆

a) $\boxed{57°17'45''}$ **b)** $\boxed{42°58'19''}$ **c)** $\boxed{143°14'22''}$

d) $\boxed{\frac{\pi}{8}}$ **e)** $\boxed{\frac{7\pi}{24}}$ **f)** $\boxed{\frac{7\pi}{8}}$

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7. ★★★ Horas: $\frac{360^\circ}{12} = \boxed{30^\circ = \frac{\pi}{6}}$. Minutos: $\frac{20}{60} \cdot 360^\circ = \boxed{120^\circ = \frac{2\pi}{3}}$. A las 4: $4 \cdot 30^\circ = \boxed{120^\circ = \frac{2\pi}{3}}$.

2 Razones trigonométricas de un ángulo agudo

8. ★☆☆

a) Pitágoras: hipotenusa 34 cm. $\sin\alpha = \boxed{\frac{8}{17}}$, $\cos\alpha = \boxed{\frac{15}{17}}$, $\operatorname{tg}\alpha = \boxed{\frac{8}{15}}$; $\sin\beta = \boxed{\frac{15}{17}}$, $\cos\beta = \boxed{\frac{8}{17}}$, $\operatorname{tg}\beta = \boxed{\frac{15}{8}}$

b) Pitágoras: cateto contiguo 8 cm. $\sin\alpha = \boxed{\frac{3}{5}}$, $\cos\alpha = \boxed{\frac{4}{5}}$, $\operatorname{tg}\alpha = \boxed{\frac{3}{4}}$; $\sin\beta = \boxed{\frac{4}{5}}$, $\cos\beta = \boxed{\frac{3}{5}}$, $\operatorname{tg}\beta = \boxed{\frac{4}{3}}$

c) Pitágoras: hipotenusa 26 cm. $\sin\alpha = \boxed{\frac{12}{13}}$, $\cos\alpha = \boxed{\frac{5}{13}}$, $\operatorname{tg}\alpha = \boxed{\frac{12}{5}}$; $\sin\beta = \boxed{\frac{5}{13}}$, $\cos\beta = \boxed{\frac{12}{13}}$, $\operatorname{tg}\beta = \boxed{\frac{5}{12}}$

d) Pitágoras: cateto opuesto 24 cm. $\sin\alpha = \boxed{\frac{24}{25}}$, $\cos\alpha = \boxed{\frac{7}{25}}$, $\operatorname{tg}\alpha = \boxed{\frac{24}{7}}$; $\sin\beta = \boxed{\frac{7}{25}}$, $\cos\beta = \boxed{\frac{24}{25}}$, $\operatorname{tg}\beta = \boxed{\frac{7}{24}}$

e) Pitágoras: cateto opuesto 21 cm. $\sin\alpha = \boxed{\frac{21}{29}}$, $\cos\alpha = \boxed{\frac{20}{29}}$, $\operatorname{tg}\alpha = \boxed{\frac{21}{20}}$; $\sin\beta = \boxed{\frac{20}{29}}$, $\cos\beta = \boxed{\frac{21}{29}}$, $\operatorname{tg}\beta = \boxed{\frac{20}{21}}$

9. ★☆☆ Pitágoras: $\overline{AD} = \sqrt{15^2 - 12^2} = 9$ cm y $\overline{BC} = \sqrt{12^2 + 16^2} = 20$ cm.

a) $\sin\hat{A} = \frac{12}{15} = \boxed{\frac{4}{5}}$, $\cos\hat{A} = \frac{9}{15} = \boxed{\frac{3}{5}}$, $\operatorname{tg}\hat{A} = \frac{12}{9} = \boxed{\frac{4}{3}}$

b) $\sin\hat{C} = \frac{12}{20} = \boxed{\frac{3}{5}}$, $\cos\hat{C} = \frac{16}{20} = \boxed{\frac{4}{5}}$, $\operatorname{tg}\hat{C} = \frac{12}{16} = \boxed{\frac{3}{4}}$

c) $\sin\widehat{ABD} = \frac{9}{15} = \boxed{\frac{3}{5}}$, $\cos\widehat{ABD} = \frac{12}{15} = \boxed{\frac{4}{5}}$, $\operatorname{tg}\widehat{ABD} = \frac{9}{12} = \boxed{\frac{3}{4}}$

d) $\sin\widehat{CBD} = \frac{16}{20} = \boxed{\frac{4}{5}}$, $\cos\widehat{CBD} = \frac{12}{20} = \boxed{\frac{3}{5}}$, $\operatorname{tg}\widehat{CBD} = \frac{16}{12} = \boxed{\frac{4}{3}}$

10. ★☆☆

α	15°	$55^\circ 20'$	$72^\circ 25' 40''$	$85{,}5^\circ$
$\sin\alpha$	$\boxed{0{,}2588}$	$\boxed{0{,}8225}$	$\boxed{0{,}9533}$	$\boxed{0{,}9969}$
$\cos\alpha$	$\boxed{0{,}9659}$	$\boxed{0{,}5688}$	$\boxed{0{,}3019}$	$\boxed{0{,}0785}$
$\operatorname{tg}\alpha$	$\boxed{0{,}2679}$	$\boxed{1{,}4460}$	$\boxed{3{,}1577}$	$\boxed{12{,}7062}$

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11. ★☆☆

a) $\alpha = \boxed{60^\circ}$ **b)** $\alpha = \approx \boxed{70^\circ 31' 44''}$ **c)** $\alpha = \approx \boxed{68^\circ 11' 55''}$

d) $\alpha = \approx \boxed{14^\circ 28' 39''}$ **e)** $\alpha = \approx \boxed{36^\circ 52' 12''}$

12. ★★☆ Medio triángulo equilátero de lado 4: catetos 2 y $\sqrt{16-4} = 2\sqrt{3}$. Medio triángulo de lado 6: catetos 3 y $\sqrt{36-9} = 3\sqrt{3}$. En los dos: $\operatorname{sin} 60^\circ = \frac{2\sqrt{3}}{4} = \frac{3\sqrt{3}}{6} = \boxed{\frac{\sqrt{3}}{2}}$, $\cos 60^\circ = \frac{2}{4} = \frac{3}{6} = \boxed{\frac{1}{2}}$, $\operatorname{tg} 60^\circ = \frac{2\sqrt{3}}{2} = \frac{3\sqrt{3}}{3} = \boxed{\sqrt{3}}$: $\boxed{\text{no dependen del lado}}$.

13. ★★☆

a) Cateto opuesto: $5 \cdot \frac{6}{5} = \boxed{6 \text{ cm}}$; hipotenusa: $\sqrt{25+36} = \boxed{\sqrt{61} \approx 7{,}81 \text{ cm}}$.

b) Hipotenusa: $\frac{24}{12/13} = \boxed{26 \text{ cm}}$; cateto: $\sqrt{26^2 - 24^2} = \boxed{10 \text{ cm}}$; $\cos = \boxed{\frac{5}{13}}$, $\operatorname{tg} = \boxed{\frac{12}{5}}$.

c) Cateto contiguo: $\frac{15}{3/4} = \boxed{20 \text{ cm}}$; hipotenusa: $\sqrt{15^2 + 20^2} = \boxed{25 \text{ cm}}$; $\operatorname{sin} = \boxed{\frac{3}{5}}$, $\cos = \boxed{\frac{4}{5}}$.

d) Los dos catetos son iguales: $x^2 + x^2 = 4 \Rightarrow x = \boxed{\sqrt{2} \text{ cm}}$ cada uno.

14. ★★★ Los catetos son $7x$ y $12x$: $49x^2 + 144x^2 = 100 \Rightarrow x = \frac{10}{\sqrt{193}}$. Catetos: $\boxed{\frac{70\sqrt{193}}{193} \approx 5{,}04 \text{ cm}}$ y $\boxed{\frac{120\sqrt{193}}{193} \approx 8{,}64 \text{ cm}}$. $\operatorname{sin} \alpha = \boxed{\frac{7\sqrt{193}}{193}}$, $\cos \alpha = \boxed{\frac{12\sqrt{193}}{193}}$.

3 Razones trigonométricas de 30°, 45° y 60°

15. ★☆☆

	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
tg	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	$\nexists$

16. ★☆☆

a) Hipotenusa: $\boxed{\frac{10\sqrt{3}}{3} \text{ cm}}$; el cateto contiguo: $\boxed{\frac{5\sqrt{3}}{3} \text{ cm}}$.

b) Hipotenusa: $\boxed{4 \text{ cm}}$; el cateto contiguo: $\boxed{2\sqrt{3} \text{ cm}}$.

c) Hipotenusa: $\boxed{7\sqrt{2} \text{ cm}}$; el cateto opuesto: $\boxed{7 \text{ cm}}$.

d) Hipotenusa: $\boxed{12 \text{ cm}}$; el cateto opuesto: $\boxed{6\sqrt{3} \text{ cm}}$.

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e) Hipotenusa: $\boxed{12\sqrt{2}\text{ cm}}$; el cateto contiguo: $\boxed{12\text{ cm}}$.

f) Cateto opuesto: $30\sin 30° = \boxed{15\text{ cm}}$; contiguo: $30\cos 30° = \boxed{15\sqrt{3}\text{ cm}}$.

17. ★☆☆

a) $\boxed{1}$ **b)** $\boxed{0}$ **c)** $\boxed{\frac{1}{2}}$

d) $\boxed{1}$ **e)** $\boxed{2}$ **f)** $\boxed{\frac{1}{2}}$

18. ★★☆ Altura: $8\sin 60° = 8\cdot\frac{\sqrt{3}}{2} = \boxed{4\sqrt{3}\approx 6{,}93\text{ cm}}$. Diagonal: $\cos 45° = \frac{5}{d} \Rightarrow d = \frac{5}{\sqrt{2}/2} = \boxed{5\sqrt{2}\approx 7{,}07\text{ cm}}$.

19. ★★☆ La apotema es la altura de un triángulo equilátero de lado 4: ap $= 4\sin 60° = \boxed{2\sqrt{3}\text{ cm}}$. Área: $6\cdot\frac{4\cdot 2\sqrt{3}}{2} = \boxed{24\sqrt{3}\approx 41{,}57\text{ cm}^2}$.

20. ★★★ Antena PB: $\overline{AB} = \frac{100}{\sin 60°} = \boxed{\frac{200\sqrt{3}}{3}\text{ m}}$, $\overline{AP} = \frac{100}{\operatorname{tg} 60°} = \frac{100\sqrt{3}}{3}$; $\overline{BC} = \frac{100}{\sin 30°} = \boxed{200\text{ m}}$, $\overline{PC} = \frac{100}{\operatorname{tg} 30°} = 100\sqrt{3}$. Antena QD (los ángulos son los de D): $\overline{DC} = \frac{75}{\cos 45°} = \boxed{75\sqrt{2}\text{ m}}$, $\overline{CQ} = 75\operatorname{tg} 45° = 75$; $\overline{DE} = \frac{75}{\cos 30°} = \boxed{50\sqrt{3}\text{ m}}$, $\overline{QE} = 75\operatorname{tg} 30° = 25\sqrt{3}$. $\overline{AE} = \frac{100\sqrt{3}}{3} + 100\sqrt{3} + 75 + 25\sqrt{3} = \boxed{75 + \frac{475\sqrt{3}}{3}\approx 349{,}24\text{ m}}$.

4 Relaciones fundamentales

21. ★☆☆

a) $\cos\alpha = \boxed{\frac{24}{25}}$, $\operatorname{tg}\alpha = \boxed{\frac{7}{24}}$

b) $\cos\alpha = \boxed{\frac{35}{37}}$, $\operatorname{tg}\alpha = \boxed{\frac{12}{35}}$

c) $\cos\alpha = \boxed{\frac{60}{61}}$, $\operatorname{tg}\alpha = \boxed{\frac{11}{60}}$

d) $\sin\alpha = \boxed{\frac{21}{29}}$, $\operatorname{tg}\alpha = \boxed{\frac{21}{20}}$

e) $\sin\alpha = \boxed{\frac{5}{13}}$, $\operatorname{tg}\alpha = \boxed{\frac{5}{12}}$

f) $\cos\alpha = \boxed{\frac{\sqrt{55}}{8}}$, $\operatorname{tg}\alpha = \boxed{\frac{3\sqrt{55}}{55}}$

g) $\sin\alpha = \boxed{\frac{2\sqrt{2}}{3}}$, $\operatorname{tg}\alpha = \boxed{2\sqrt{2}}$

h) $\cos\alpha = \boxed{\frac{7\sqrt{19}}{34}}$, $\operatorname{tg}\alpha = \boxed{\frac{15\sqrt{19}}{133}}$

i) $\sin\alpha = \boxed{\frac{\sqrt{21}}{5}}$, $\operatorname{tg}\alpha = \boxed{\frac{\sqrt{21}}{2}}$

j) $\cos\alpha = \boxed{\frac{\sqrt{33}}{7}}$, $\operatorname{tg}\alpha = \boxed{\frac{4\sqrt{33}}{33}}$

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22. ★★☆

a) $\sin\alpha = \boxed{\frac{5}{13}}$, $\cos\alpha = \boxed{\frac{12}{13}}$

b) $\sin\alpha = \boxed{\frac{3}{5}}$, $\cos\alpha = \boxed{\frac{4}{5}}$

c) $\sin\alpha = \boxed{\frac{5\sqrt{34}}{34}}$, $\cos\alpha = \boxed{\frac{3\sqrt{34}}{34}}$

d) $\sin\alpha = \boxed{\frac{5\sqrt{194}}{194}}$, $\cos\alpha = \boxed{\frac{13\sqrt{194}}{194}}$

e) $\sin\alpha = \boxed{\frac{5\sqrt{89}}{89}}$, $\cos\alpha = \boxed{\frac{8\sqrt{89}}{89}}$

f) $\sin\alpha = \boxed{\frac{3\sqrt{10}}{10}}$, $\cos\alpha = \boxed{\frac{\sqrt{10}}{10}}$

g) $\sin\alpha = \boxed{\frac{\sqrt{2}}{2}}$, $\cos\alpha = \boxed{\frac{\sqrt{2}}{2}}$

23. ★☆☆

a) $\sin\alpha \approx \boxed{0{,}9907}$, $\operatorname{tg}\alpha \approx \boxed{7{,}3010}$

b) $\sin\alpha \approx \boxed{0{,}3027}$, $\operatorname{tg}\alpha \approx \boxed{0{,}3175}$

c) $\sin\alpha \approx \boxed{0{,}9581}$, $\operatorname{tg}\alpha \approx \boxed{3{,}3454}$

d) $\cos\alpha \approx \boxed{0{,}9832}$, $\operatorname{tg}\alpha \approx \boxed{0{,}1858}$

e) $\cos\alpha \approx \boxed{0{,}2992}$, $\operatorname{tg}\alpha \approx \boxed{3{,}1895}$

f) $\cos\alpha \approx \boxed{0{,}8915}$, $\operatorname{tg}\alpha \approx \boxed{0{,}5083}$

24. ★★☆

a) $\frac{49}{625} + \frac{121}{625} = \frac{170}{625} \neq 1$: no hay ninguno

b) $\frac{49}{169} + \frac{144}{169} = \frac{193}{169} \neq 1$: no hay ninguno

c) $\frac{5}{25} + \frac{20}{25} = 1$: sí que hay

d) $\frac{9}{25} + \frac{16}{25} = 1$: sí que hay

e) $\cos\alpha = \sqrt{1 - \frac{49}{121}} = \frac{6\sqrt{2}}{11}$ y $\frac{\sin\alpha}{\cos\alpha} = \frac{7}{6\sqrt{2}} \neq \frac{6}{11}$: no hay ninguno

f) $1^2 + 1 = 2 \neq \frac{169}{25} = \frac{1}{\cos^2\alpha}$: no hay ninguno

g) $\frac{3}{9} + 1 = \frac{4}{3} = \frac{1}{3/4}$: sí que hay

h) $\frac{49}{25} + 1 = \frac{74}{25} \neq \frac{25}{9}$: no hay ninguno

25. ★★☆

a) $0{,}4663^2 + 1 \approx 1{,}2174$, $\frac{1}{0{,}1736^2} \approx 33{,}1818$: no

b) $0{,}3640^2 + 1 \approx 1{,}1325$, $\frac{1}{0{,}9397^2} \approx 1{,}1325$: sí

c) $0{,}1485^2 + 0{,}2588^2 \approx 0{,}0890$: no

d) $0{,}9325^2 + 1 \approx 1{,}8696$, $\frac{1}{0{,}7313^2} \approx 1{,}8699$: sí

e) $0{,}2754^2 + 0{,}6691^2 \approx 0{,}5235$: no

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26. ★★★

a) Dividimos $\sin^2\alpha + \cos^2\alpha = 1$ entre $\sin^2\alpha$: $1 + \dfrac{\cos^2\alpha}{\sin^2\alpha} = \dfrac{1}{\sin^2\alpha}$, y $\dfrac{\cos^2\alpha}{\sin^2\alpha} = \dfrac{1}{\operatorname{tg}^2\alpha}$.

b) $(1+\operatorname{tg}^2\alpha)\cos^2\alpha = \cos^2\alpha + \dfrac{\sin^2\alpha}{\cos^2\alpha}\cos^2\alpha = \cos^2\alpha + \sin^2\alpha = 1.$

c) $\sin\alpha\cos\alpha\dfrac{\sin\alpha}{\cos\alpha} + \cos^2\alpha = \sin^2\alpha + \cos^2\alpha = 1.$

5 Resolución de triángulos rectángulos

27. ★☆☆

a) $\hat{C} = \boxed{60°}$, $a = \frac{4}{\sin 30°} = \boxed{8\text{ m}}$, $c = \frac{4}{\operatorname{tg} 30°} = \boxed{4\sqrt{3} \approx 6{,}93\text{ m}}$

b) $c = \sqrt{7^2 - 5^2} = \boxed{2\sqrt{6} \approx 4{,}9\text{ m}}$, $\sin\hat{B} = \frac{5}{7} \Rightarrow \hat{B} \approx \boxed{45°35'5''}$, $\hat{C} \approx \boxed{44°24'55''}$

c) $\hat{C} = \boxed{50°}$, $b = 10\sin 40° \approx \boxed{6{,}43\text{ m}}$, $c = 10\cos 40° \approx \boxed{7{,}66\text{ m}}$

d) $a = \frac{5}{0{,}25} = \boxed{20\text{ cm}}$, $b = \sqrt{20^2 - 5^2} = \boxed{5\sqrt{15} \approx 19{,}36\text{ cm}}$, $\hat{C} \approx \boxed{14°28'39''}$, $\hat{B} \approx \boxed{75°31'21''}$

28. ★☆☆

a) $h = 18\sin 62° \approx \boxed{15{,}89\text{ cm}}$

b) $h = 28\sin 35° \approx \boxed{16{,}06\text{ cm}}$

29. ★☆☆ $\cos\alpha = \frac{1{,}2}{3} = 0{,}4 \Rightarrow \alpha \approx \boxed{66°25'19''}$

30. ★☆☆ $\operatorname{tg}\alpha = \frac{20}{25} = 0{,}8 \Rightarrow \alpha \approx \boxed{38°39'35''}$

31. ★★☆ La altura lo divide en dos triángulos rectángulos de catetos 9 y 10: $\operatorname{tg}\hat{B} = \frac{10}{9} \Rightarrow$ ángulos iguales $\approx \boxed{48°0'46''}$; ángulo desigual $\approx 180° - 2\cdot 48°0'46'' = \boxed{83°58'28''}$.

32. ★★☆ Desnivel: $1265 - 785 = 480$ m. Pendiente: $\frac{480}{3000} = \boxed{0{,}16 = 16\,\%}$. Ángulo: $\operatorname{tg}^{-1} 0{,}16 \approx \boxed{9°5'25''}$.

33. ★★☆ La distancia recorrida es la hipotenusa: $\sin\alpha = \frac{2\,160}{18\,410} \Rightarrow \alpha \approx \boxed{6°44'16''}$.

34. ★★☆ El triángulo es isósceles: la altura divide el ángulo en dos de $25°$. $r = 2\cdot 12\sin 25° \approx \boxed{10{,}14\text{ cm}}$.

35. ★★☆ El ángulo recto está en C: $\overline{AC} = \frac{120}{\operatorname{tg} 37°} \approx 159{,}25$ m, $\overline{AB} = \frac{120}{\sin 37°} \approx 199{,}4$ m. Diferencia: $\approx \boxed{40{,}15\text{ m}}$.

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6 Problemas de trigonometría

36. ★★☆ $h = x\,\mathrm{tg}\,35° = (x+200)\,\mathrm{tg}\,20° \Rightarrow x = \dfrac{200\,\mathrm{tg}\,20°}{\mathrm{tg}\,35° - \mathrm{tg}\,20°} \approx 216{,}5$ m; $h \approx \boxed{151{,}59\text{ m}}$.

37. ★★☆ Desde el punto más próximo el ángulo es mayor: $h = x\,\mathrm{tg}\,46° = (x+100)\,\mathrm{tg}\,33° \Rightarrow x \approx 168{,}19$ m; $h \approx \boxed{174{,}16\text{ m}}$.

38. ★★☆ $h = x\,\mathrm{tg}\,76° = (x+480)\,\mathrm{tg}\,45° \Rightarrow x \approx 159{,}43$ m, $h \approx 639{,}43$ m sobre los puntos de observación. Altitud de la cima: $1200 + h \approx \boxed{1839{,}43\text{ m}}$.

39. ★★☆ $h = x\,\mathrm{tg}\,50° = (x+5)\,\mathrm{tg}\,40° \Rightarrow x \approx 11{,}9$ m; $h \approx \boxed{14{,}18\text{ m}}$.

40. ★★☆ Las ciudades están a lados opuestos: $h = x\,\mathrm{tg}\,36° = (75-x)\,\mathrm{tg}\,12° \Rightarrow x \approx 16{,}98$ km; $h \approx \boxed{12{,}33\text{ km}}$. Distancias: $\frac{h}{\sin 36°} \approx \boxed{20{,}98\text{ km}}$ de A y $\frac{h}{\sin 12°} \approx \boxed{59{,}32\text{ km}}$ de B.

41. ★★☆ $h = x\,\mathrm{tg}\,60° = (140-x)\,\mathrm{tg}\,40° \Rightarrow x \approx 45{,}69$ cm; la lámpara queda a $h \approx \boxed{79{,}14\text{ cm}}$ del techo. Cuerda: $\frac{h}{\sin 60°} + \frac{h}{\sin 40°} \approx \boxed{214{,}49\text{ cm}}$.

42. ★★☆ $h = x\,\mathrm{tg}\,60° = (126-x)\,\mathrm{tg}\,30° \Rightarrow \sqrt{3}\,x = \frac{126-x}{\sqrt{3}} \Rightarrow x = 31{,}5$ m. Altura: $\boxed{\dfrac{63\sqrt{3}}{2} \approx 54{,}56\text{ m}}$.

Cables: $\overline{AB} = \frac{31{,}5}{\cos 60°} = \boxed{63\text{ m}}$, $\overline{BC} = \frac{94{,}5}{\cos 30°} = \boxed{63\sqrt{3} \approx 109{,}12\text{ m}}$. $\widehat{ABC} = 180° - 60° - 30° = \boxed{90°}$.

43. ★★☆ $10\cos 45° + 10\cos 30° = 5\sqrt{2} + 5\sqrt{3} \approx \boxed{15{,}73\text{ m}}$

44. ★★☆ $25\sin 30° + 30\sin 50° \approx 12{,}5 + 22{,}98 \approx \boxed{35{,}48\text{ m}}$ (el tramo horizontal no cuenta).

45. ★★☆

a) $\mathrm{tg}\,36° = \frac{5}{\mathrm{ap}} \Rightarrow \mathrm{ap} = \frac{5}{\mathrm{tg}\,36°} \approx \boxed{6{,}88\text{ cm}}$

b) Son 5 triángulos isósceles de lados 6 y 6 y ángulo $72°$: altura $6\sin 72°$; $S = 5 \cdot \frac{6 \cdot 6\sin 72°}{2} \approx \boxed{85{,}6\text{ cm}^2}$

46. ★★☆

a) $h = \frac{2{,}5}{\mathrm{tg}\,27{,}5°} \approx \boxed{4{,}8\text{ cm}}$; $S = \frac{5h}{2} \approx \boxed{12{,}01\text{ cm}^2}$

b) Lados iguales: $\frac{17}{\sin 25°} \approx 40{,}23$ cm; $P \approx \boxed{114{,}45\text{ cm}}$; $h = \frac{17}{\mathrm{tg}\,25°} \approx 36{,}46$ cm, $S \approx \boxed{619{,}76\text{ cm}^2}$

47. ★★☆

a) $S = 12 \cdot 12\sin 43° \approx \boxed{98{,}21\text{ cm}^2}$

b) Las diagonales se cortan perpendicularmente en el punto medio: $\mathrm{tg}\,\frac{\alpha}{2} = \frac{6}{4} \Rightarrow \alpha \approx \boxed{112°37'12''}$ y $\beta \approx \boxed{67°22'48''}$; lado $\sqrt{6^2+4^2} = 2\sqrt{13}$, $P = \boxed{8\sqrt{13} \approx 28{,}84\text{ cm}}$; $S = \frac{12 \cdot 8}{2} = \boxed{48\text{ cm}^2}$

48. ★★★ Trazando la altura sobre el lado de 15 cm: $h = 10\sin 35° \approx 5{,}74$, proyección $10\cos 35° \approx 8{,}19$. Diagonal menor: $\sqrt{h^2 + (15-8{,}19)^2} \approx \boxed{8{,}9\text{ cm}}$. Área: $15 \cdot h \approx \boxed{86{,}04\text{ cm}^2}$.

49. ★★★ Trazamos la altura desde A sobre BC: $h = 245\sin 57° \approx 205{,}47$ m, $\overline{CH} = 245\cos 57° \approx 133{,}44$ m, $\overline{HB} \approx 524{,}56$ m. $\overline{AB} = \sqrt{h^2 + \overline{HB}^2} \approx \boxed{563{,}37\text{ m}}$.

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50. ★★★ La portería va de 27,5 m a 32,5 m de altura sobre la línea de banda y A está a 50 m de la línea de fondo. Ángulo de AB con la línea de banda: $\text{tg}^{-1}\frac{27{,}5}{50} \approx 28°48'39''$; ángulo de AC: $\text{tg}^{-1}\frac{32{,}5}{50} \approx 33°1'26''$. $\alpha \approx \boxed{4°12'47''}$.

7 Razones trigonométricas de un ángulo cualquiera

51. ★☆☆

α	$\sin\alpha$	$\cos\alpha$	$\text{tg}\,\alpha$	**Cuadrante**
$120°$	$\boxed{0{,}8660}$	$\boxed{-0{,}5000}$	$\boxed{-1{,}7321}$	$\boxed{\text{II}}$
$220°$	$\boxed{-0{,}6428}$	$\boxed{-0{,}7660}$	$\boxed{0{,}8391}$	$\boxed{\text{III}}$
$135°$	$\boxed{0{,}7071}$	$\boxed{-0{,}7071}$	$\boxed{-1{,}0000}$	$\boxed{\text{II}}$
$300°$	$\boxed{-0{,}8660}$	$\boxed{0{,}5000}$	$\boxed{-1{,}7321}$	$\boxed{\text{IV}}$
$240°$	$\boxed{-0{,}8660}$	$\boxed{-0{,}5000}$	$\boxed{1{,}7321}$	$\boxed{\text{III}}$
$75°$	$\boxed{0{,}9659}$	$\boxed{0{,}2588}$	$\boxed{3{,}7321}$	$\boxed{\text{I}}$

52. ★☆☆

a) $\boxed{\text{III o IV}}$ **b)** $\boxed{\text{II o III}}$ **c)** $\boxed{\text{II o IV}}$

53. ★☆☆

a) $\boxed{-1}$ **b)** $\boxed{1}$ **c)** $\boxed{1}$ **d)** $\boxed{1}$

54. ★★☆

a) $\sin\alpha = \sqrt{1-0{,}49} \approx \boxed{0{,}7141}$, $\text{tg}\,\alpha \approx \boxed{1{,}0202}$

b) $\frac{1}{\cos^2\beta} = 9+1 = 10$, $\cos\beta = \boxed{-\frac{\sqrt{10}}{10}}$ (negativo en el II), $\sin\beta = \text{tg}\,\beta\cdot\cos\beta = \boxed{\frac{3\sqrt{10}}{10}}$

c) $\cos\gamma = -\sqrt{1-\frac{9}{25}} = \boxed{-\frac{4}{5}}$ (negativo en el III), $\text{tg}\,\gamma = \frac{-3/5}{-4/5} = \boxed{\frac{3}{4}}$

d) $\sin\delta = -\sqrt{1-\frac{25}{169}} = \boxed{-\frac{12}{13}}$ (negativo en el IV), $\text{tg}\,\delta = \boxed{-\frac{12}{5}}$

55. ★★☆ $\cos^2\alpha = 1-0{,}34^2 = 0{,}8844 \Rightarrow \cos\alpha = \pm 0{,}9404$. Como el seno es positivo, α puede ser del I o del II cuadrante: I: $\cos\alpha \approx \boxed{0{,}9404}$, $\text{tg}\,\alpha \approx \boxed{0{,}3615}$ ($\alpha \approx 19°52'37''$); II: $\cos\alpha \approx \boxed{-0{,}9404}$, $\text{tg}\,\alpha \approx \boxed{-0{,}3615}$ ($\alpha \approx 160°7'23''$).

56. ★★☆

a) $\boxed{210°}$, $\boxed{330°}$ **b)** $\boxed{135°}$, $\boxed{315°}$

c) $\boxed{36°52'12''}$, $\boxed{323°7'48''}$ **d)** $\boxed{64°9'29''}$, $\boxed{115°50'31''}$

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57. ★★☆ Seno negativo y coseno positivo: $\boxed{\text{cuarto cuadrante}}$. $\beta = 360° - 30° = \boxed{330°}$.

58. ★☆☆

a) $1750° = 4 \cdot 360° + 310°$: $\boxed{310°}$

b) $1920° = 5 \cdot 360° + 120°$: $\boxed{120°}$

c) $-800° = -3 \cdot 360° + 280°$: $\boxed{280°}$

d) $3825° = 10 \cdot 360° + 225°$: $\boxed{225°}$

8 Reducción al primer cuadrante

59. ★☆☆

a) $135° = 180° - 45°$: $\sin = \boxed{\frac{\sqrt{2}}{2}}$, $\cos = \boxed{-\frac{\sqrt{2}}{2}}$, $\text{tg} = \boxed{-1}$

b) $240° = 180° + 60°$: $\sin = \boxed{-\frac{\sqrt{3}}{2}}$, $\cos = \boxed{-\frac{1}{2}}$, $\text{tg} = \boxed{\sqrt{3}}$

c) $-30° = -360° + 330°$; $330° = 360° - 30°$: $\sin = \boxed{-\frac{1}{2}}$, $\cos = \boxed{\frac{\sqrt{3}}{2}}$, $\text{tg} = \boxed{-\frac{\sqrt{3}}{3}}$

60. ★★☆

a) $155° = 180° - 25°$: $\sin 155° = \boxed{\sin 25°}$, $\cos 155° = \boxed{-\cos 25°}$, $\text{tg}\, 155° = \boxed{-\text{tg}\, 25°}$

b) $240° = 180° + 60°$: $\sin = \boxed{-\frac{\sqrt{3}}{2}}$, $\cos = \boxed{-\frac{1}{2}}$, $\text{tg} = \boxed{\sqrt{3}}$

c) $390° = 360° + 30°$: $\sin = \boxed{\frac{1}{2}}$, $\cos = \boxed{\frac{\sqrt{3}}{2}}$, $\text{tg} = \boxed{\frac{\sqrt{3}}{3}}$

d) $225° = 180° + 45°$: $\sin = \boxed{-\frac{\sqrt{2}}{2}}$, $\cos = \boxed{-\frac{\sqrt{2}}{2}}$, $\text{tg} = \boxed{1}$

e) $1920° = 5 \cdot 360° + 120°$; $120° = 180° - 60°$: $\sin = \boxed{\frac{\sqrt{3}}{2}}$, $\cos = \boxed{-\frac{1}{2}}$, $\text{tg} = \boxed{-\sqrt{3}}$

f) $-30° = -360° + 330°$; $330° = 360° - 30°$: $\sin = \boxed{-\frac{1}{2}}$, $\cos = \boxed{\frac{\sqrt{3}}{2}}$, $\text{tg} = \boxed{-\frac{\sqrt{3}}{3}}$

61. ★★☆

a) $-40° = -360° + 320°$; $320° = 360° - 40°$: $\sin(-40°) = \boxed{-\sin 40°}$, $\cos(-40°) = \boxed{\cos 40°}$, $\text{tg}(-40°) = \boxed{-\text{tg}\, 40°}$

b) $385° = 360° + 25°$: $\sin 385° = \boxed{\sin 25°}$, $\cos 385° = \boxed{\cos 25°}$, $\text{tg}\, 385° = \boxed{\text{tg}\, 25°}$

c) $-185° = -360° + 175°$; $175° = 180° - 5°$: $\sin(-185°) = \boxed{\sin 5°}$, $\cos(-185°) = \boxed{-\cos 5°}$, $\text{tg}(-185°) = \boxed{-\text{tg}\, 5°}$

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d) $835° = 2 \cdot 360° + 115°$; $115° = 180° - 65°$: $\operatorname{sen} 835° = \boxed{\operatorname{sen} 65°}$, $\cos 835° = \boxed{-\cos 65°}$, $\operatorname{tg} 835° = \boxed{-\operatorname{tg} 65°}$

e) $934° = 2 \cdot 360° + 214°$; $214° = 180° + 34°$: $\operatorname{sen} 934° = \boxed{-\operatorname{sen} 34°}$, $\cos 934° = \boxed{-\cos 34°}$, $\operatorname{tg} 934° = \boxed{\operatorname{tg} 34°}$

62. ★★☆

a) $145°$: $\operatorname{sen} \approx \boxed{0{,}57}$, $\cos \approx \boxed{-0{,}82}$, $\operatorname{tg} \approx \boxed{-0{,}7}$

b) $215°$: $\operatorname{sen} \approx \boxed{-0{,}57}$, $\cos \approx \boxed{-0{,}82}$, $\operatorname{tg} \approx \boxed{0{,}7}$

c) $325°$: $\operatorname{sen} \approx \boxed{-0{,}57}$, $\cos \approx \boxed{0{,}82}$, $\operatorname{tg} \approx \boxed{-0{,}7}$

d) $55°$: $\operatorname{sen} \approx \boxed{0{,}82}$, $\cos \approx \boxed{0{,}57}$, $\operatorname{tg} \approx \boxed{1{,}43}$

e) $-35°$: $\operatorname{sen} \approx \boxed{-0{,}57}$, $\cos \approx \boxed{0{,}82}$, $\operatorname{tg} \approx \boxed{-0{,}7}$

f) $395°$: $\operatorname{sen} \approx \boxed{0{,}57}$, $\cos \approx \boxed{0{,}82}$, $\operatorname{tg} \approx \boxed{0{,}7}$

63. ★★★

a) $120° = 180° - 60°$: $\operatorname{sen} 120° = \operatorname{sen} 60° = \boxed{\cos 30°}$, $\cos 120° = -\cos 60° = \boxed{-\operatorname{sen} 30°}$, $\operatorname{tg} 120° = -\operatorname{tg} 60° = \boxed{-\dfrac{1}{\operatorname{tg} 30°}}$

b) $430° = 360° + 70°$: $\operatorname{sen} 430° = \operatorname{sen} 70° = \boxed{\cos 20°}$, $\cos 430° = \boxed{\operatorname{sen} 20°}$, $\operatorname{tg} 430° = \boxed{\dfrac{1}{\operatorname{tg} 20°}}$

c) $-800° = -3 \cdot 360° + 280°$ y $280° = 360° - 80°$: $\operatorname{sen}(-800°) = -\operatorname{sen} 80° = \boxed{-\cos 10°}$, $\cos(-800°) = \cos 80° = \boxed{\operatorname{sen} 10°}$, $\operatorname{tg}(-800°) = -\operatorname{tg} 80° = \boxed{-\dfrac{1}{\operatorname{tg} 10°}}$

d) $\operatorname{sen} 180° = \boxed{\operatorname{sen} 0° = 0}$, $\cos 180° = \boxed{-\cos 0° = -1}$, $\operatorname{tg} 180° = \boxed{\operatorname{tg} 0° = 0}$

64. ★★★

$\operatorname{sen} \alpha - \operatorname{sen} \alpha + \cos \alpha - \cos \alpha = \boxed{0}$

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